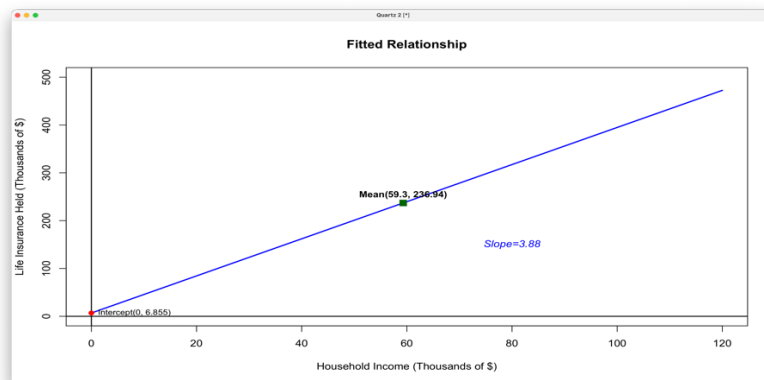


18.a

sample mean of insurance = $6.855 + 3.880 \cdot 59.3 = 236.939$



18.b

point estimate: $\Delta \text{insurance held} = 3.880 \cdot 1 = 3.880$ (thousands)

95%CI: $b_2 \pm t_{0.025, 18} \cdot \text{se}(b_2) \Rightarrow (3.6447, 4.1153)$

On average, we have 95% confidence that when income increases \$1000, the average insurance held will increase \$3880.

18.c

point estimate: $y = 6.855 + 3.880 \cdot 100 = 394.855$

$\text{var}(y) = 7.383^2 + 0.112^2 \cdot 100 + 2 \cdot 100 \cdot (-.746) = 30.74869$

99%CI: $y \pm t_{0.005, 18} \cdot \text{se}(y) \Rightarrow (378.8936, 410.8164)$

18.d

$H_0: \beta_2 = 5, H_a: \beta_2 \neq 5, df = 18$

$t = (3.880 - 5) / 0.112 = -10$

rejection region: $|t| > t_{0.025, 18} = 2.100922$

Since $t = -10 < -t_{0.025, 18} = -2.100922$, Reject H_0 , accept H_a . There is statistically significant that the insurance held increase is lower than \$5,000.

18.e

$$H_0: \beta_2 = 1, H_a: \beta_2 > 1, df = 18$$

$$t = (3.880 - 1) / 0.112 = 25.71429$$

$$\text{rejection region: } t > t_{0.01, 18} = 2.55238$$

Since $t = 25.71429 > t_{0.01, 18} = 2.55238$, Reject H_0 , accept H_a . There is statistically significant that the slope is larger than one.

23.a

$$\text{marginal effect} = dE(\text{price})/d\text{SQFT} = 2\alpha_2 \text{SQFT} = 40\alpha_2 = 7.38076$$

$$H_0: 40\alpha_2 = 13, H_a: 40\alpha_2 > 13, df = 498$$

$$t = (7.38076 - 13) / 40\text{se}(\alpha_2) = -26.72875$$

$$\text{rejection region: } t > t_{0.05, 498} = 1.647919$$

$$p\text{-value} = 1$$

Since $p\text{-value} = 1$, non-reject H_0 . There is no significant proof that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is more than \$13000.

23.b

$$\text{marginal effect} = dE(\text{price})/d\text{SQFT} = 2\alpha_2 \text{SQFT} = 80\alpha_2 = 14.76152$$

$$H_0: 80\alpha_2 = 13, H_a: 80\alpha_2 > 13, df = 498$$

$$t = (14.76152 - 13) / 40\text{se}(\alpha_2) = 4.189467$$

$$\text{rejection region: } t > t_{0.05, 498} = 1.647919$$

$$p\text{-value} = 1.65395e-05$$

Since $p\text{-value} = 1.65395e-05$, Reject H_0 , accept H_a . There is significant proof that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is more than \$13000.

23.c

$$E(\text{price}|\text{SQFT}) = \alpha_1 + \alpha_2 \text{SQFT}^2 = 167.3735$$

95% CI: (158.0481, 176.6988)

On average, the expected price for a house of 2000 square feet house is \$167.3735 (thousand). We have 95% confidence that the average price for house will fall between \$158.0481 and \$176.6988.

23.d

sample mean=163.3333 falls in 95% C.I. It's compatible with (c).

31.a

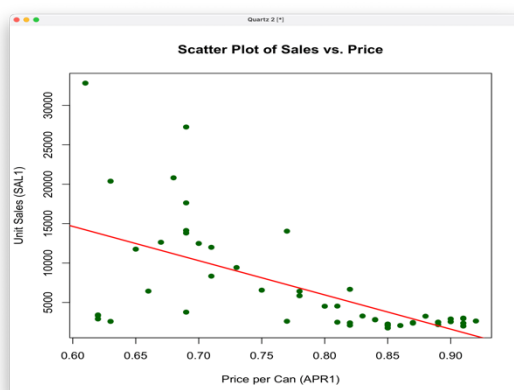
There is substantial variation in weekly sales, while the variation in price is relatively smaller but discrete.

```
> print(sal_summary)
      Mean      Min      Max      SD
6718.712 1762.000 32820.000 6915.083
> print(apr_summary)
      Mean      Min      Max      SD
0.78250000 0.61000000 0.92000000 0.09803711
```



31.b

Inverse relationship, yes. By the Law of Demand, price and quantity of demand move inversely.



31.c

point estimate=-434.45

95% CI=>(-592.2897,-276.6049)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	40714.22	6196.42	6.571	2.82e-08 ***
PRICE1	-434.45	78.58	-5.528	1.17e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom

Multiple R-squared: 0.3794, Adjusted R-squared: 0.367

F-statistic: 30.56 on 1 and 50 DF, p-value: 1.172e-06

```
>
> conf_interval_c <- confint(model_c, "PRICE1", level = 0.95)
> print(conf_interval_c)
      2.5 %      97.5 %
PRICE1 -592.2897 -276.6049
```

31.d

	fit	lwr	upr
$y=40714.22+(-434.45*70)=10302.72$	1 10302.9	8624.934	11980.87

90% C.I.=>(8624.934 ,11980.87)

When the price of a can is 70 cents, we have 90% confidence that the sales will fall between \$8624.934 and \$11980.87.

31.e

point estimate=-434.45*78.25/6718.71=-5.059825

se=78.58*78.25/6718.71=0.9152451

95% CI: -5.059825± $t_{0.025,50}$ *se=> (-6.898149 , -3.221501)

Sales are elastic. When the price rises 1%, sales will fall by -5. 059825%.

It makes sense because it is common food and has so many suppliers. If one supplier increases its price, the sales must fall a lot.

31.f

H_0 : elasticity=-3, H_a : elasticity $\neq$ -3, df=50, $\alpha=.1$

$t=(-5.059825-3)/0.9152451=-2.250572$

rejection region: $|t|>t_{0.05,50}=1.675905$

p-value=0.02884204

Since p-value=0.02884204<0.05, Reject H_0 , accept H_a . There is significant proof that the elasticity not equal to -3. The elasticity is even stronger.